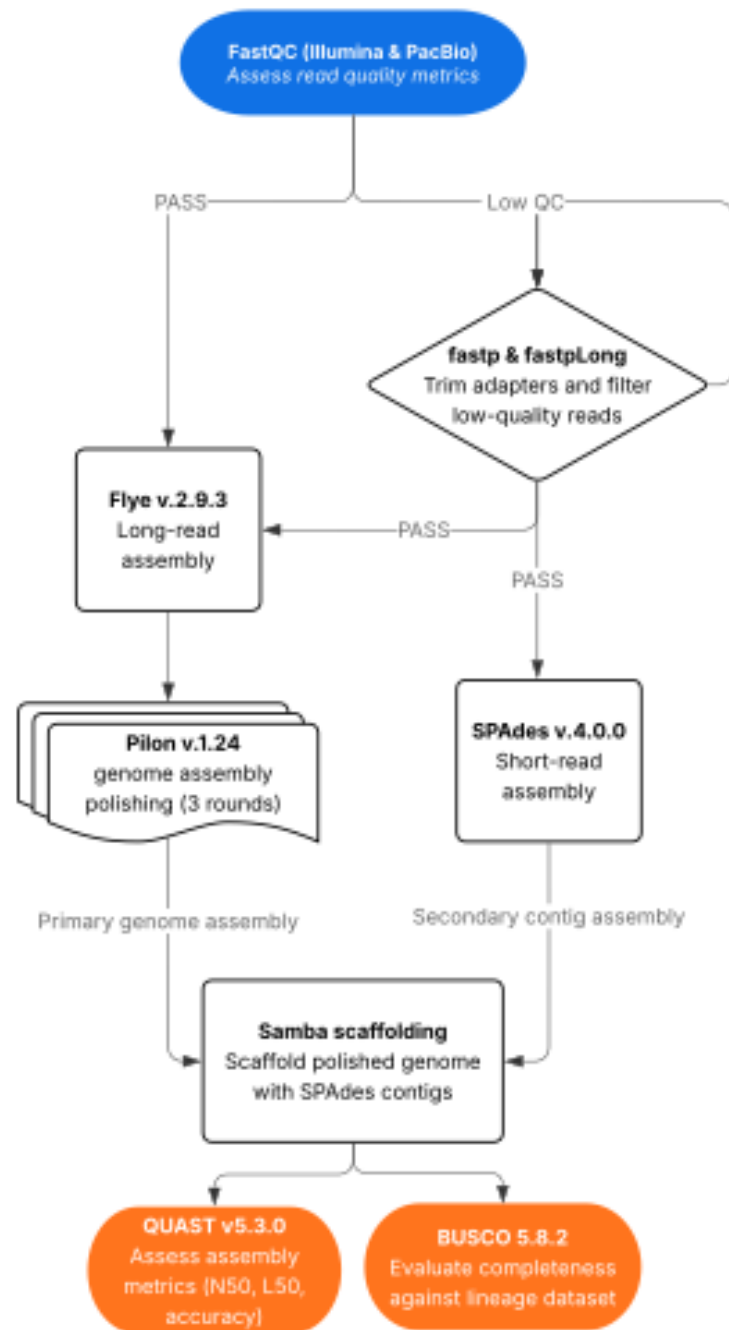


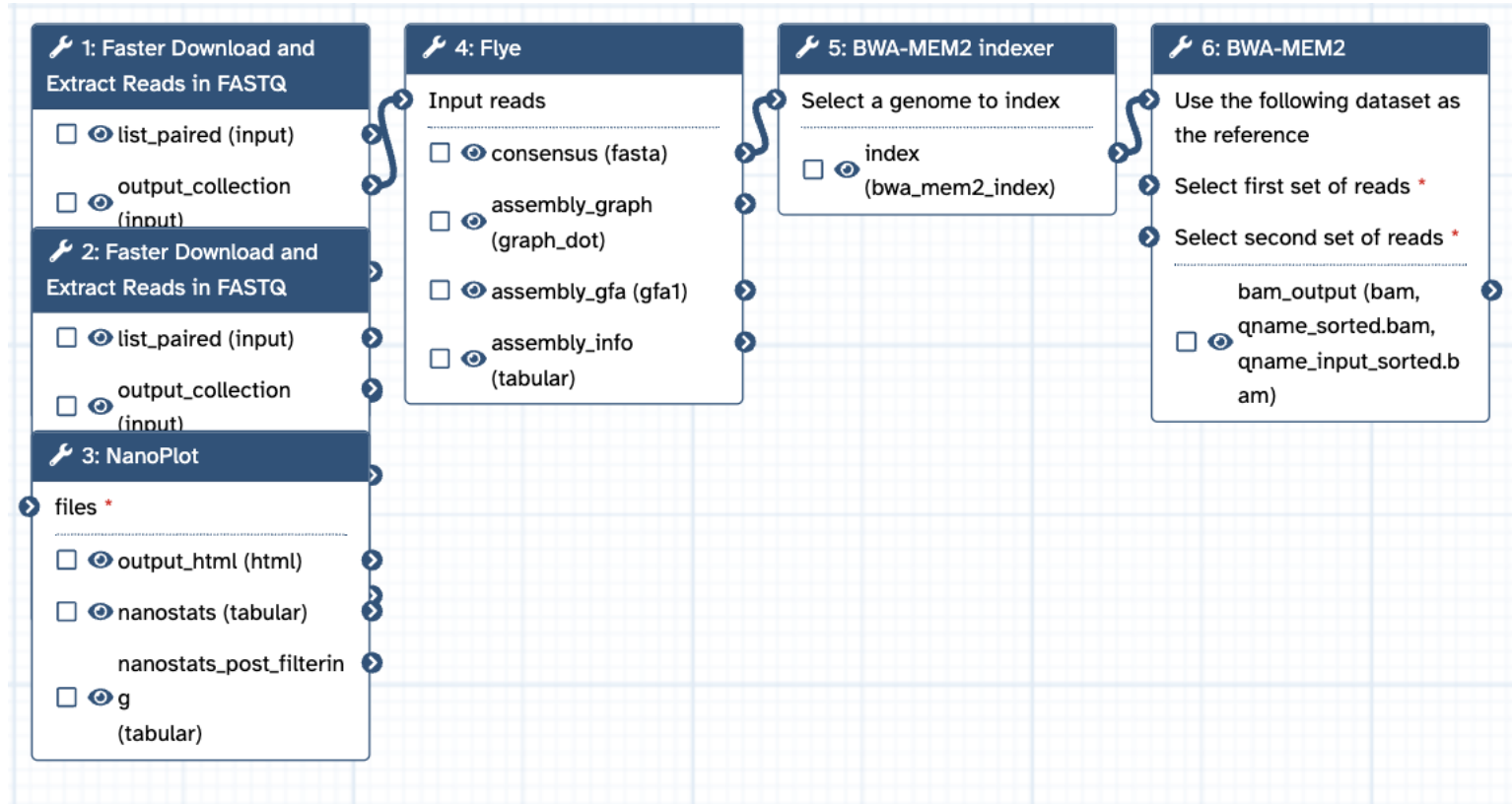
Genome assembly



- * *P. coatneyi* (ncbi + our data)
- * *Haemoproteus majoris* (ncbi)

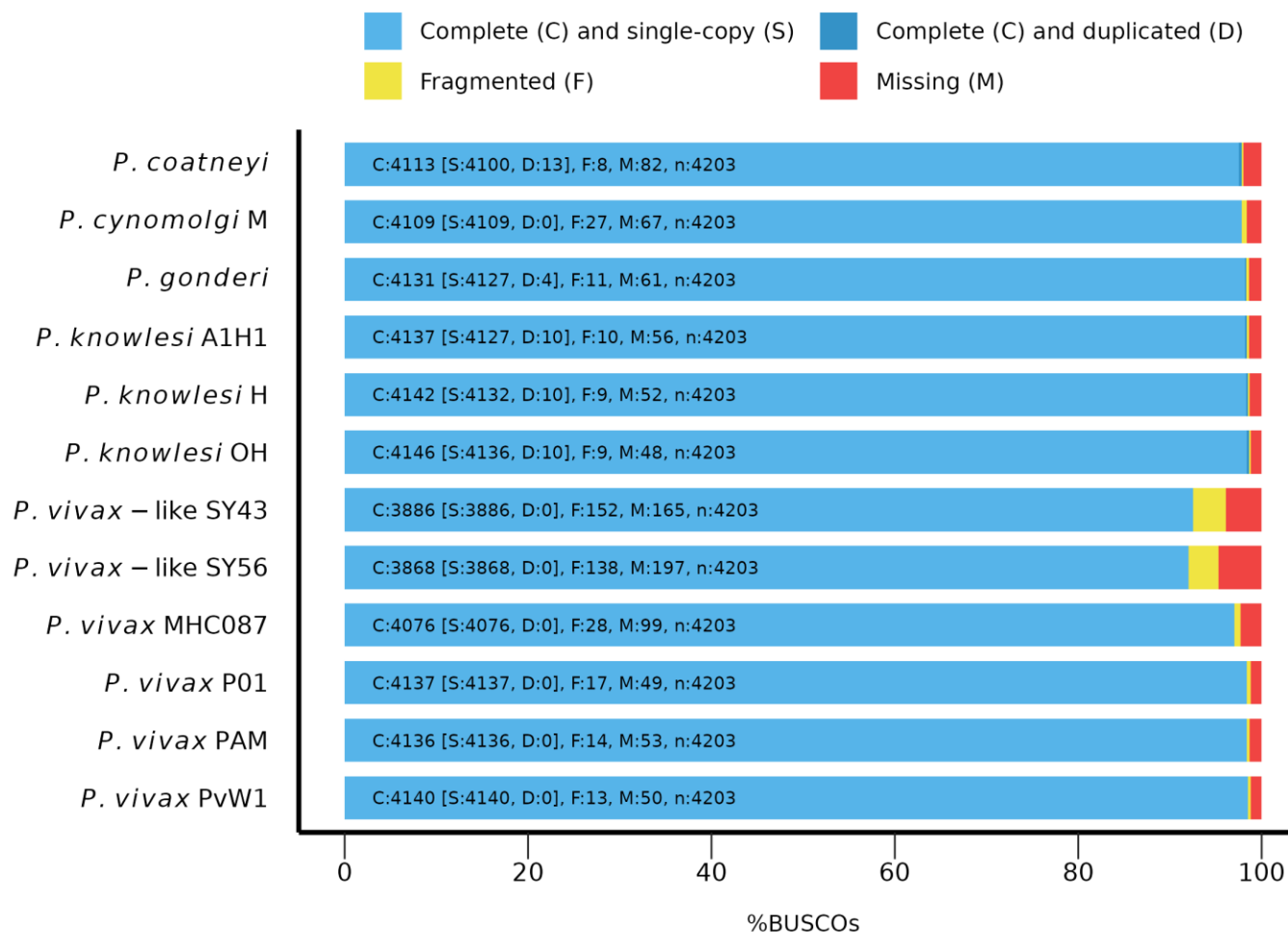
Long-read assembly (Flye)

Preprocessing short-reads



For polishing a genome assembly, Galaxy currently lacks available (medaka) or functional (pilon) tools

BUSCO Assessment Results



**Busco**

Assess genome assembly and annotation completeness
(Galaxy Version 5.8.0+galaxy2)

Tool Parameters

! Please provide a value for this option.

Sequences to analyse * required



accepted formats ▼

No datasets with fasta or fasta.gz elements available

Can be an assembled genome or transcriptome (DNA), or protein sequences from an annotated gene set.

Cached database with lineage *

Busco v5 Lineage Datasets ▼

Mode

Genome assemblies (DNA) ▼

(--mode)

Select a gene predictor

Miniprot ▼

In the case of a prokaryotic genome, Prodigal is the default gene predictor

Auto-detect or select lineage?

Auto-detect ▼

Let BUSCO determine the best lineage automatically, or select from known lineage

auto-lineage group *

All taxonomic groups (--auto-lineage) ▼

Taxonomic group to run with auto-lineage.

Do you want to display the results from the generic lineage

☒ No

BUSCO generates files for the parent lineage (generic) and the final selected lineage (specific). Generally, the lineage to select for your assessments should be the most specific lineage available. The generic lineage datasets for the domains archaea, bacteria and eukaryota.

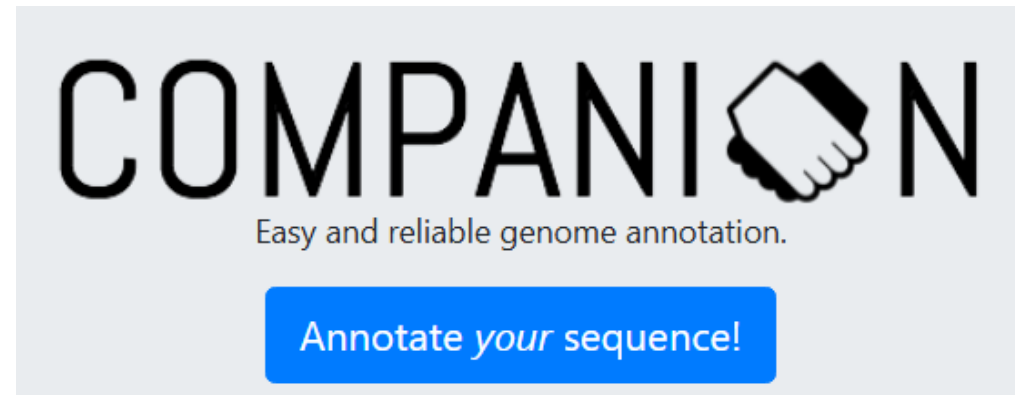
Which outputs should be generated - optional

Short summary text ▼

switch to column select ▼

I searched for additional options, but only the default, non-specialized databases are available.

* *P. coatneyi*



What species are offered as references?

We import annotated parasite genomes from external sources like [VEuPathDB](#) and preprocess them for use as references. Currently, we are offering [434 reference species](#), which can be seen [here](#).

2. Install *Companion*

Execute these commands in the directory you want to keep your *Companion* work in. Do this as a normal user, i.e. not as root or using sudo. Use a name that is meaningful to you in place of `<my-companion-project>`

```
curl -L -o companion-master.zip https://github.com/sii-companion/companion/archive/master.zip && \
  unzip companion-master.zip && \
  mv companion-master <my-companion-project>
docker pull uofgiii/companion
```

* *Haemoproteus majoris*

 **InterProScan** functional annotation (Galaxy Version 5.59-91.0+galaxy3) ☆ ▼  ▶ Run Tool

Tool Parameters

! Please provide a value for this option.

Protein FASTA File * *required*

   ...

No datasets with fasta elements available

accepted formats ▼

(--input)

Type of the input sequences *

Protein ▼

(--seqtype)

InterProScan database *

InterProScan 5.59-91.0 ▼

Applications to run - optional

Search for options

Aa .*

Unselected (0)

Select all →

Selected (18)

← Deselect all

▲

TIGRFAM: protein families based on hidden Markov models (HMMs)

FunFam: Prediction of functional annotations for novel, uncharacterized sequences.

SFLD: a database of protein families based on hidden Markov models (HMMs)

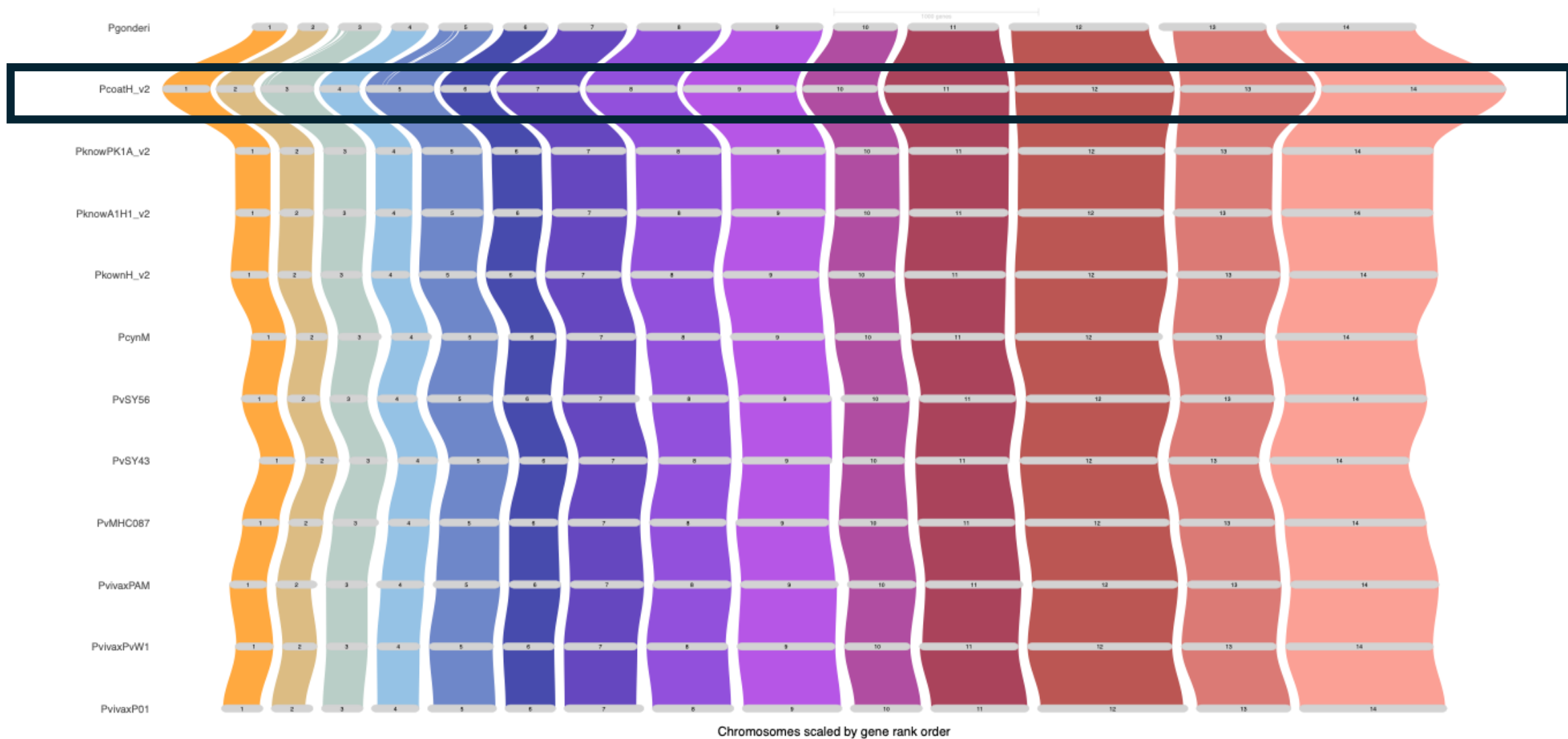
SUPERFAMILY: database of structural and functional annotation for all proteins and genomes

▼ PANTHER: Protein ANalysis THrough Evolutionary Relationships

▼

[interproscan-5.76-107.0 Latest](#)
Oct 16, 2025

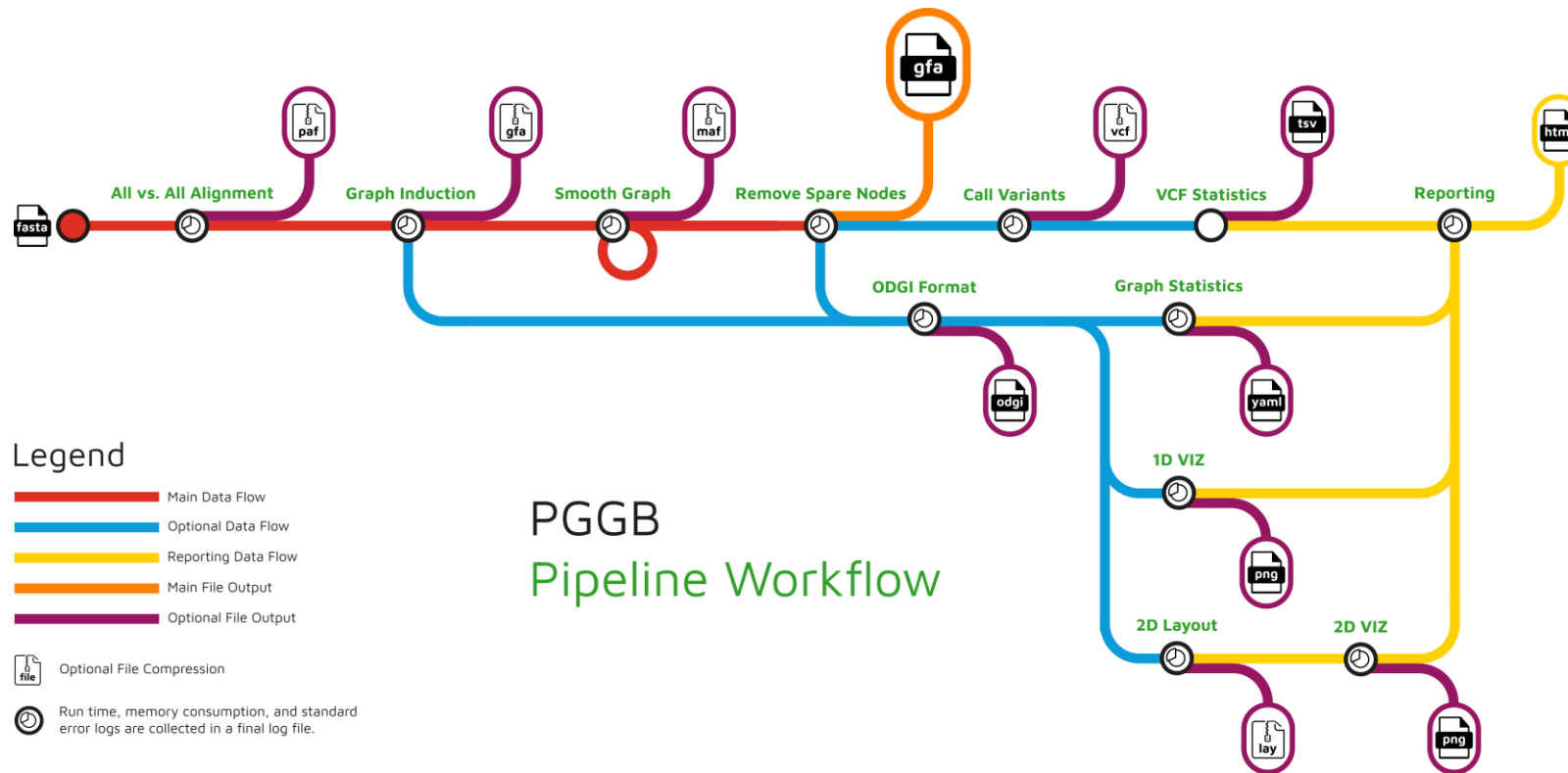
High structural conservation



GENESPACE: an R package for synteny- and orthology-constrained comparative genomics.

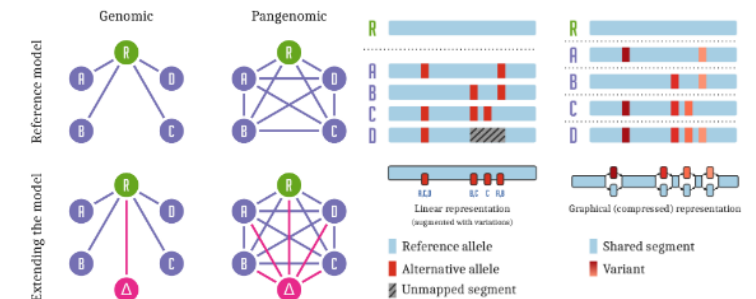
Which other genomic features show comparable conservation (centromeres, promoters, etc.)?

Can we reliably detect species-specific and clade-restricted genomic regions? Multigene family enrichment?



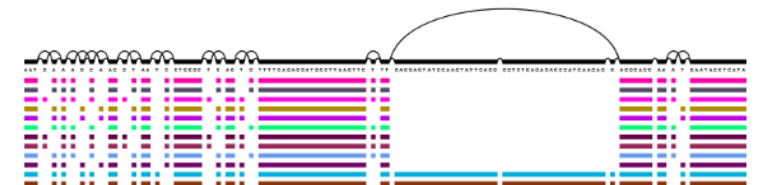
Pangenomic methods

Standard approaches to genome inference and analysis relate sequences to a single linear reference genome. This is efficient but has a fundamental problem: Differences from this reference are hard to observe and describe in a coherent way. Variation and sequence are separated.



Pangenomic methods allow us to relate all genomes or sequences in our analysis directly to each other. Sequence and variation are combined into a coherent data structure. This practice is still new, and research into ways to design, implement, and apply this model is ongoing. However, there is a growing consensus around best practices. Many methods work on an augmented sequence graph model and use a handful of common data formats for input and output.

The *variation graph* data model describes the all-to-all alignment of many sequences (genomes or genes for instance) as walks through a graph whose nodes are labeled with DNA sequences:





Welcome to the interactive Galaxy IPython Notebook.

This is a Markdown Cell

These cells can be used to describe the thought process driving the analysis for sharing efficiently your results

```
In [3]: print "This is a command output displayed"  
## This is a command line cell to run your analysis.  
## Compatible with several languages
```

This is a command output displayed

```
In [ ]:
```




Filter files by name

Name	Modified
/	
centromere_extr...	14d ago
chr01_p85_s5kb	13d ago
chr02_p85_s5kb	14d ago
galaxy_inputs	14d ago
outputs	14d ago
chr01.fasta	14d ago
chr01.fasta.fai	14d ago
chr01.fasta.seqk...	14d ago
chr02.fasta	14d ago
chr02.fasta.fai	14d ago
chr02.fasta.seqk...	14d ago
ipython_galaxy_...	14d ago
PlasmodiumPGG...	13d ago
run_pggb.sh	14d ago

PlasmodiumPGGBTuto.ipyn X

run_pggb.sh X +

Code Open in...

```
[ ]: !conda install bioconda::pggb -y
```

```
[ ]: !conda install bioconda::seqkit -y
```

```
[9]: !seqkit
!pggb
```

SeqKit -- a cross-platform and ultrafast toolkit for FASTA/Q file manipulation

Version: 2.10.1

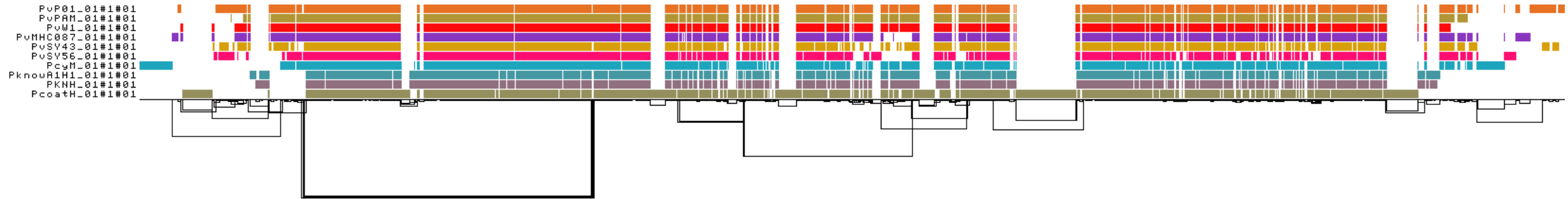
Author: Wei Shen <shenwei356@gmail.com>

Documents : <http://bioinf.shenwei.me/seqkit>
 Source code: <https://github.com/shenwei356/seqkit>
 Please cite: <https://doi.org/10.1002/imt2.191>

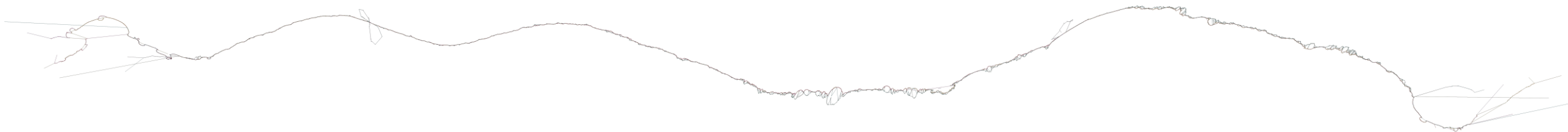
Seqkit utilizes the pgzip (<https://github.com/klauspost/pgzip>) package to read and write gzip file, and the outputted gzip file would be slightly larger than files generated by GNU gzip.

Seqkit writes gzip files very fast, much faster than the multi-threaded pigz,

PGGB results chr 01

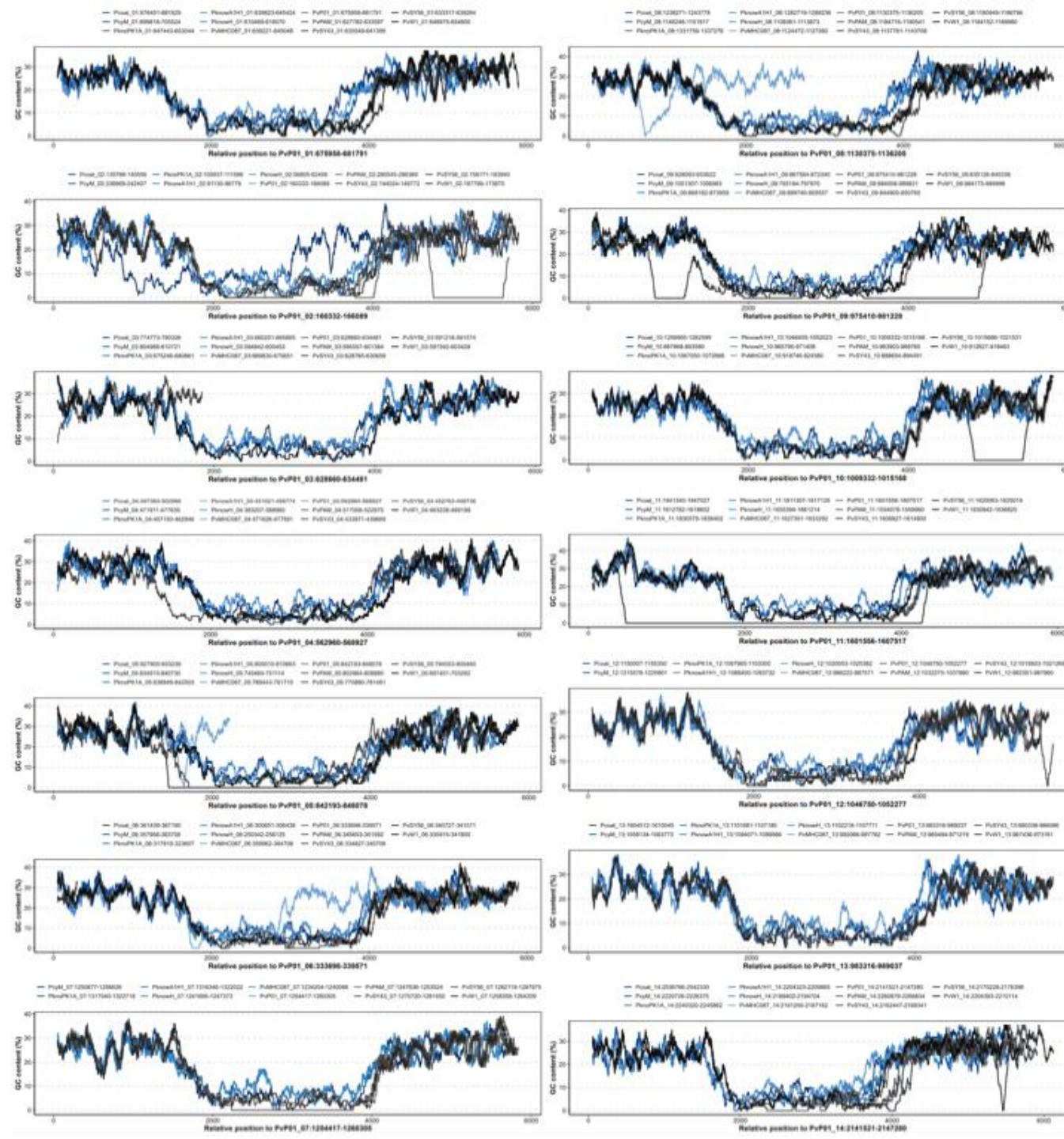


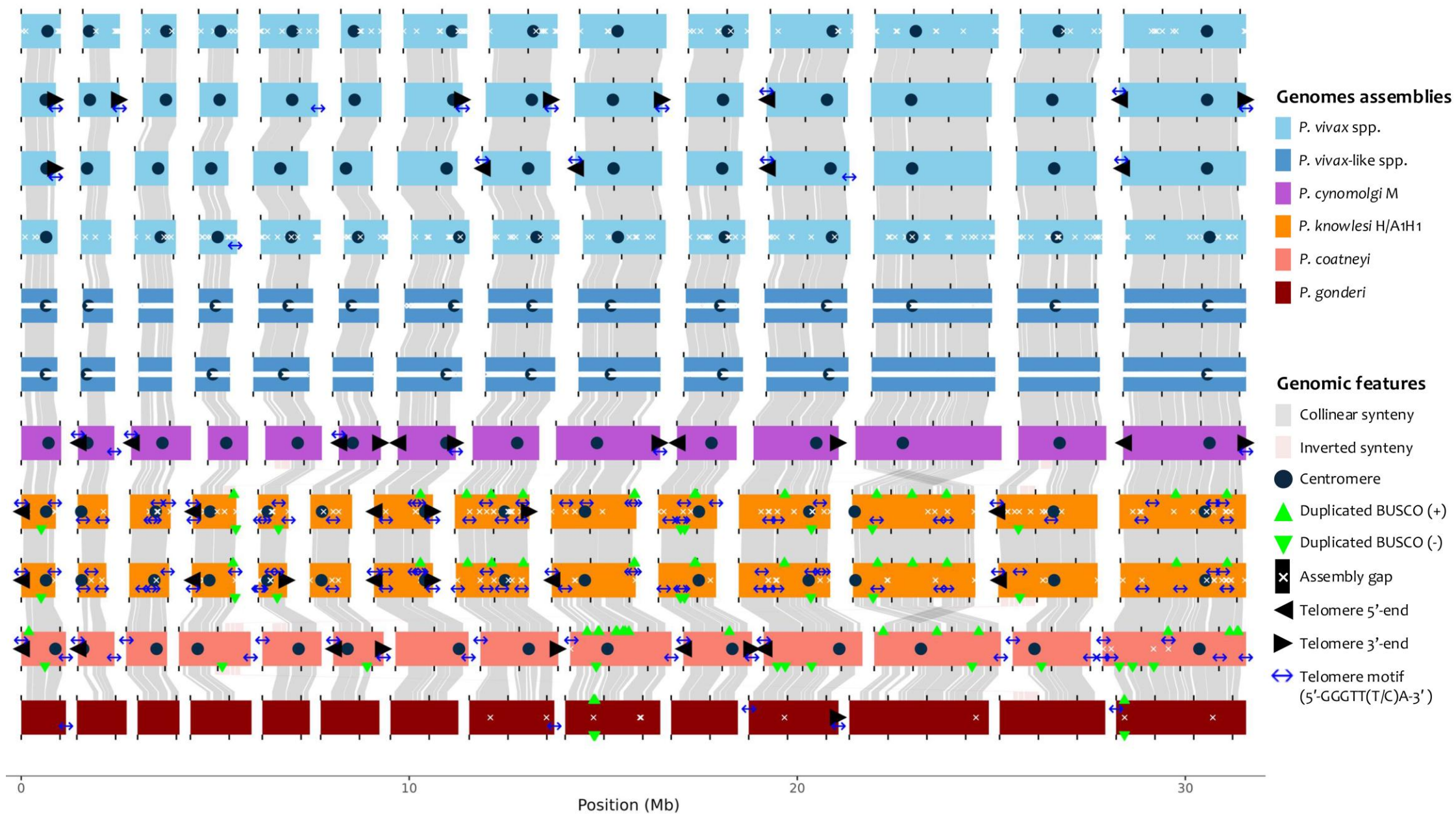
1D alignment plot



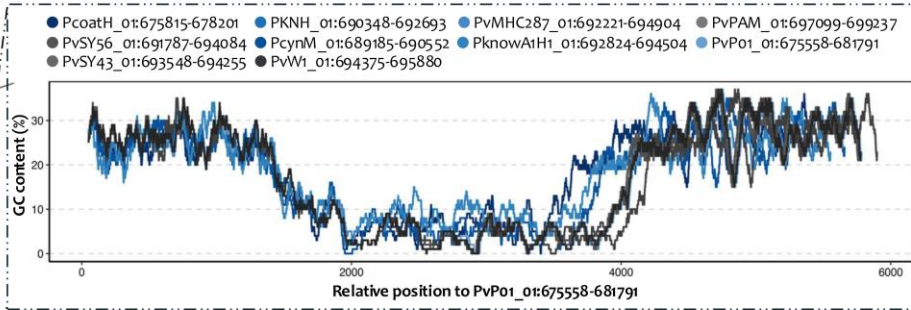
2D alignment plot

Centromeres
1.8-2.2kb
5-7% GC
refSeq: *PvivaxPo1*

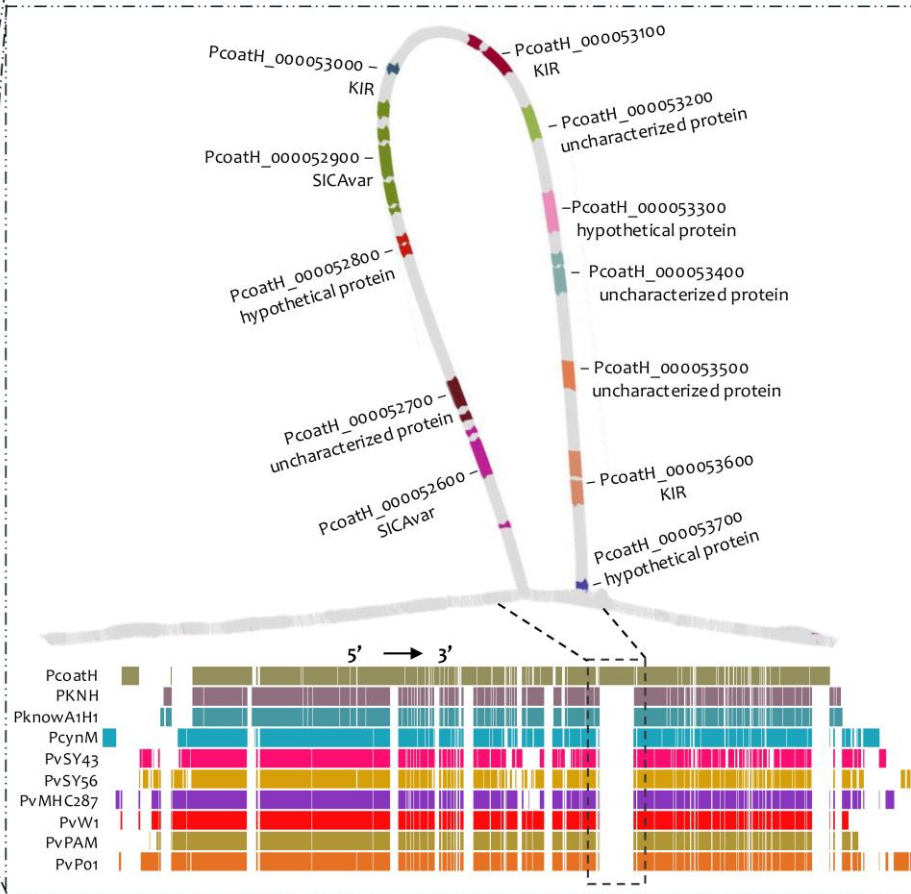




Chro1
5'
↓
3'



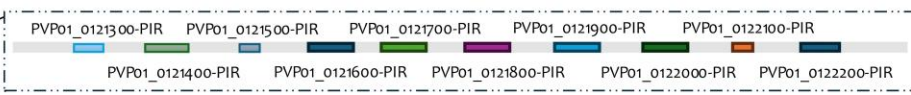
R plot



Multigene family enrichment

PGGB (2D)

Bandage software (2D)



PGGB (1D)

Bandage software (2D)

[GGGTTTA]₄₅₀ - *P. vivax* PAM

tidd software